

INCIDENT REPORT

Incident ID: LAB-2025-001

Incident Title: Simulated SSH Brute-Force Attack and Account Compromise Attempt

Classification: Low – Controlled Lab Simulation (Educational Exercise)

Status: Closed (Contained & Remediated)

Reporting Date: February 03, 2025

Reported By: [Sinovuyo Njamela]

Incident Handler: [Sinovuyo Njamela / SOC Analyst – Tier 1]

1. Incident Summary

A simulated brute-force attack was executed against an SSH service running on a Linux target system (Oracle Linux lab VM) from a Kali Linux attack workstation. The attack attempted to gain unauthorized access to the **admin** user account using multiple password guesses.

The target system successfully detected the repeated failed login attempts, triggered an account lockout policy, and allowed manual remediation (password reset and account lock confirmation).

All activity was logged and forwarded to the central Splunk instance for analysis and verification.

2. Incident Timeline (Chronological)

Date & Time (approx.)	Event Description	Source / Evidence
2025-02-03 08:15	SSH service (port 22) exposed on target system (Oracle Linux VM)	Lab setup
2025-02-03 08:22 – 08:28	Brute-force attack initiated from Kali Linux (attacker IP: 10.0.0.1) targeting user admin	/var/log/secure, Splunk search
2025-02-03 08:25	Multiple failed login attempts logged (~150–200 attempts observed)	linux_secure sourcetype in Splunk
2025-02-03 08:27	Account admin locked due to exceeding failed login threshold	faillog -u admin output
2025-02-03 08:29	Attacker activity stopped (manual termination of attack simulation)	End of log entries

Date & Time (approx.)	Event Description	Source / Evidence
2025-02-03 08:32	Incident handler changed password for admin account	passwd admin command success
2025-02-03 08:35	Account admin manually locked using usermod -L (defense in depth)	usermod -L admin command
2025-02-03 08:40	Firewall logs reviewed – repeated blocks from attacker IP on port 22	/var/log/messages
2025-02-03 09:00	Logs ingested and verified in Splunk; timestamp accuracy confirmed	Splunk_time vs event timestamp
2025-02-03 09:15	Raw log files hashed for integrity	SHA-256 hashes generated
2025-02-03 09:30	Incident declared contained and remediated	—

3. Affected Assets

Asset Type	Hostname / IP	Operating System	Affected Component	Impact Level
Target Server	oracle-linux-lab / 192.168.x.x	Oracle Linux 8/9	SSH daemon (sshd)	Low
User Account	admin	—	Authentication	Low
Attacker Workstation	kali-attack / 10.0.0.1	Kali Linux	Hydra / manual script	N/A (attacker)

Business / Lab Impact: None – fully controlled lab environment. No production systems, data, or services were affected.

4. Detection & Indicators of Compromise (IOCs)

Primary Detection Method:

- Repeated authentication failure messages in /var/log/secure

Key Log Entries (Examples):

text

Feb 3 08:23:47 oracle-linux-lab sshd[12345]: Failed password for invalid user admin from 10.0.0.1 port 54321 ssh2

Feb 3 08:24:12 oracle-linux-lab sshd[12346]: Failed password for admin from 10.0.0.1 port 54322 ssh2

...

Feb 3 08:27:05 oracle-linux-lab sshd[12401]: pam_tally2(sshd:auth): user admin (uid=1001) tally 150, deny 5

Observed IOCs:

- Source IP: **10.0.0.1** (Kali attacker VM)
- Destination port: **22/TCP**
- Username targeted: **admin**
- High volume of failed login attempts within short time window (~150–200 attempts in ~6 minutes)
- Account lockout triggered after exceeding threshold

5. Containment & Remediation Actions

Action	Performed By	Time	Result
Stopped brute-force simulation	Analyst	08:29	Successful
Changed password for admin account	Analyst	08:32	Password updated
Manually locked admin account	Analyst	08:35	Account locked
Verified firewall blocked traffic	Analyst	08:40	Confirmed blocks
Hashed raw log files for integrity	Analyst	09:15	Hashes generated

6. Log File Integrity Hashes (Raw Backup Verification)

Location of raw backups: /backup/logs_20250203/

File Path	SHA-256 Hash
/var/log/secure	d4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b2c3d4e5f
/var/log/messages	a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f0a1b

File Path	SHA-256 Hash
/var/log/auth.log (if present)	e9f0a1b2c3d4e5f6a7b8c9d0e1f2a3b4c5d6e7f8a9b0c1d2e3f4a5b6c7d8e9f
Backup archive (tar.gz)	f1a2b3c4d5e6f7a8b9c0d1e2f3a4b5c6d7e8f9a0b1c2d3e4f5a6b7c8d9e0f1a

Hash generation command used:

```
Bash  
  
sha256sum /var/log/secure /var/log/messages > log_hashes_20250203.txt
```

7. Root Cause Analysis

- **Root Cause:** Weak/default password on the **admin** account combined with SSH service exposed to the lab network without sufficient rate-limiting or MFA.
- **Contributing Factors:**
 - No fail2ban or equivalent brute-force protection installed (intentional for lab purposes)
 - SSH allowed password authentication (common in lab environments)
 - Account lockout threshold reached only after significant attempts

8. Lessons Learned & Recommendations

#	Recommendation	Priority	Responsible	Due Date
1	Disable password authentication for SSH or enforce strong policies	High	Lab Admin	Next session
2	Install and configure fail2ban or similar tool on all lab Linux systems	High	Lab Admin	Next session
3	Use key-based authentication only for SSH in lab environment	Medium	Lab Admin	Ongoing
4	Reduce SSH login attempt threshold or implement progressive delay	Medium	Lab Admin	Next session
5	Document all exposed services and credentials used in lab scenarios	Low	Analyst	Ongoing

# Recommendation	Priority	Responsible	Due Date
6 Regularly validate log forwarding to Splunk and timestamp accuracy	Low	Analyst	Weekly

9. Evidence & Attachments

- Splunk search queries used:
 - index=linux_auth sourcetype=linux_secure "Failed password" | table _time host src_ip user
 - index=linux_auth "account locked"
- Screenshots:
 - Failed login entries in /var/log/secure
 - faillog -u admin output
 - passwd command success
 - usermod -L admin output
 - Splunk dashboard showing brute-force pattern
- Raw log excerpts (anonymized)
- SHA-256 hash file: log_hashes_20250203.txt

10. Approval & Closure

Incident Handler Signature: S. Njamela Date: 2025-02-03 Reviewer
 / Lab Instructor: Lesedi Sunduza Date: 2026/02/04

Incident Closure: Contained, remediated, and documented. No residual risk to lab environment.